Predicting Residential Solar Adoption

Across U.S. Counties

Colin Segura • Springboard Data Science Career Track • Capstone Two — Final Report

The Problem

U.S. residential solar has grown **23×** since 2010 — from 30,000 to over 700,000 installs per year.

But growth is deeply uneven. Two counties with nearly identical sunshine, income, and home ownership can differ by **10×** in installations per capita.

The industry has no clean explanation for why.

Installers

Spend 15–20% of revenue on CAC. Everyone targets the same high-income zip codes.

Utilities

Under-forecasting distributed solar creates voltage issues on feeders and unplanned upgrades.

State energy offices

Spend billions on incentives. Can't cleanly separate policy effect from income or price.

Data & Method

THREE PUBLIC DATA SOURCES

LBNL Tracking the Sun

3M+ residential PV installs
with date, size, location, cost

→ Modeling target

U.S. Census ACS

County-level income, housing units,
owner-occupied count, year built

→ Features + denominator

EIA Open Data

State-level residential retail
electricity price by year

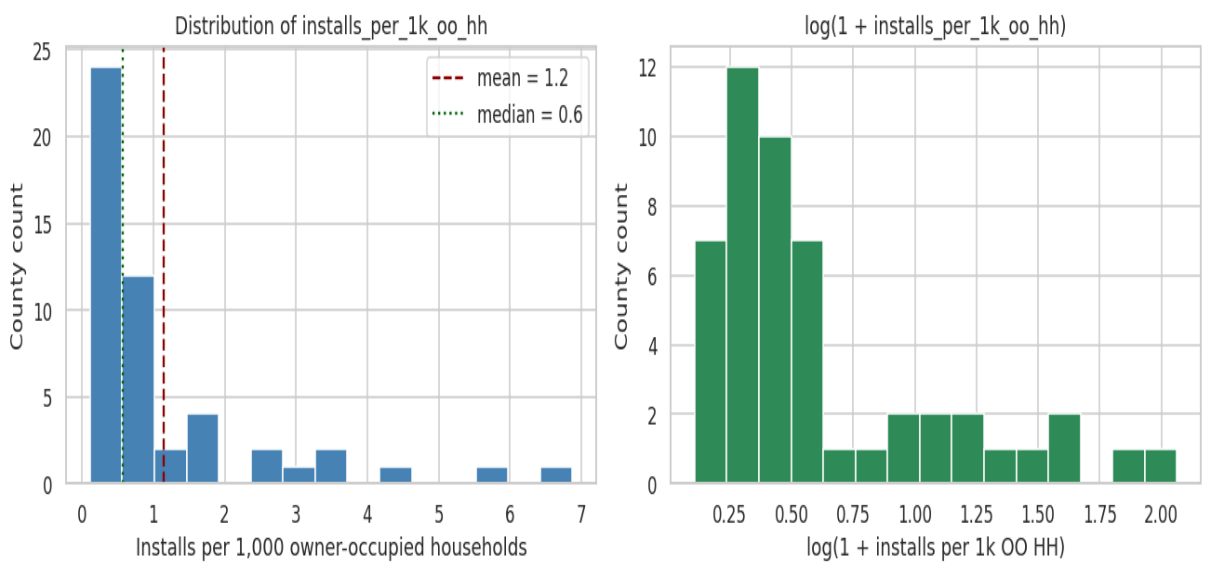
→ Key economic feature

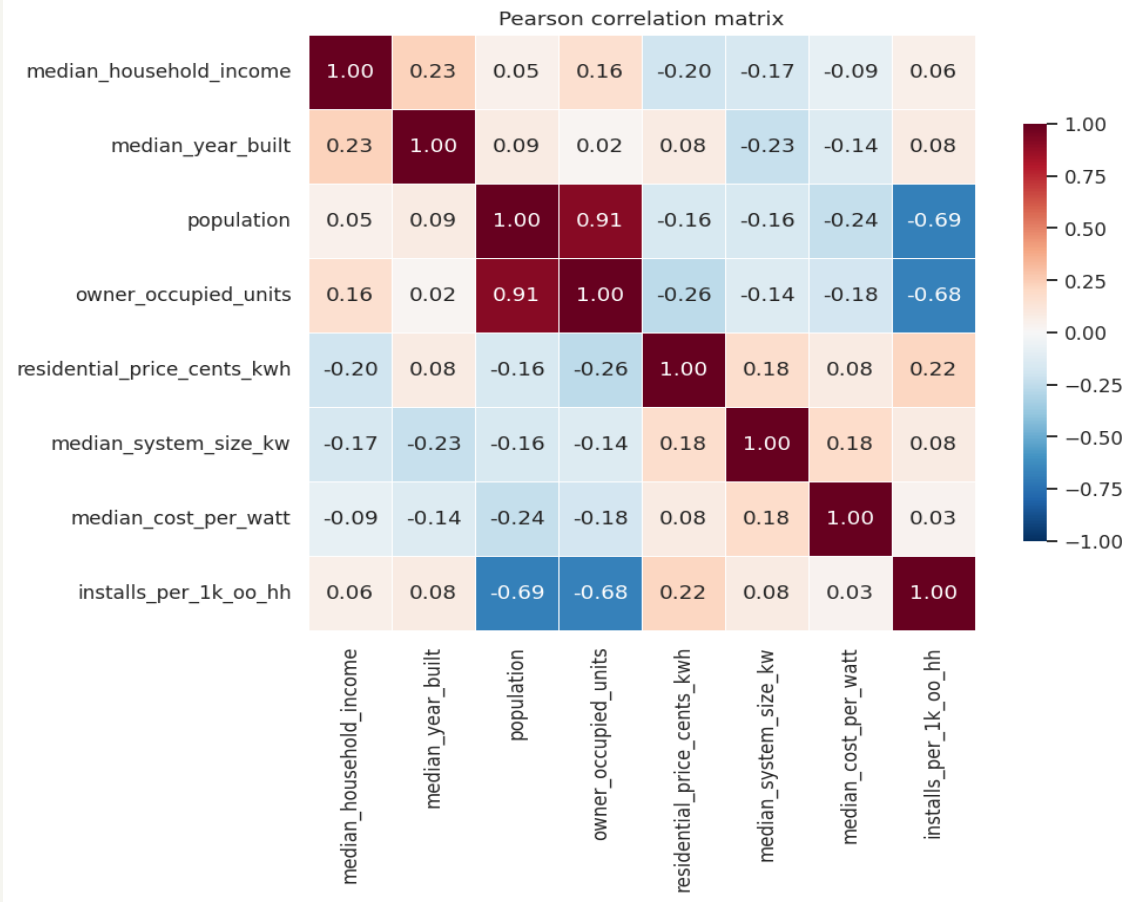
METHOD

1. Data wrangling: merge on county FIPS, aggregate to county-level target
2. EDA: distribution, correlations, subgroup analysis
3. Preprocessing: impute missing, engineer features, standardize, 80/20 train/test split
4. Modeling: compare 4 regressors with 5-fold CV, hyperparameter tune, select on RMSE

Key EDA finding: population dominates

Contrary to the industry heuristic, county population — not electricity price or income — is the strongest single correlate of per-capita adoption. And the sign is negative.

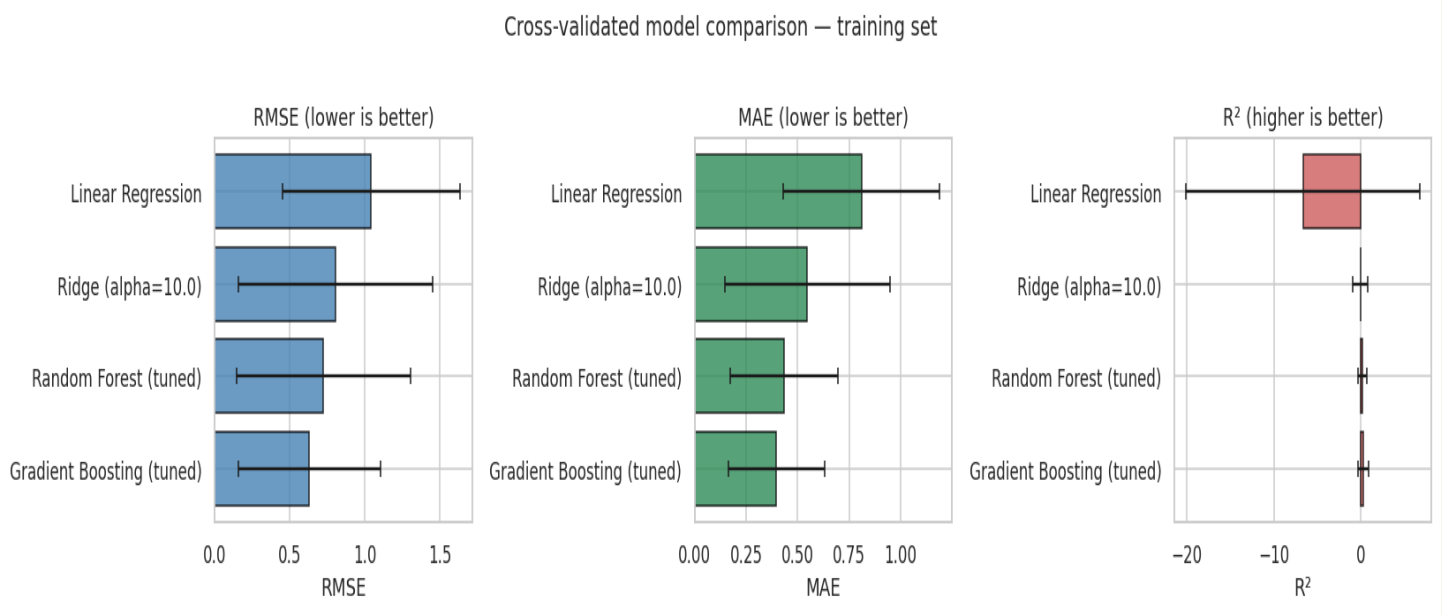




TAKEAWAYS

- Target is right-skewed (skewness = 2.4) → log-transform for linear models
- Population r = **−0.69** with target (strong negative)
- Electricity price r = +0.22, Spearman ρ = +0.31
- Income r ≈ **+0.06** → no linear relationship

Four models compared, one winner



WINNER

Gradient Boosting

Regressor (tuned)

CV RMSE **0.63**

CV R² **+0.28**

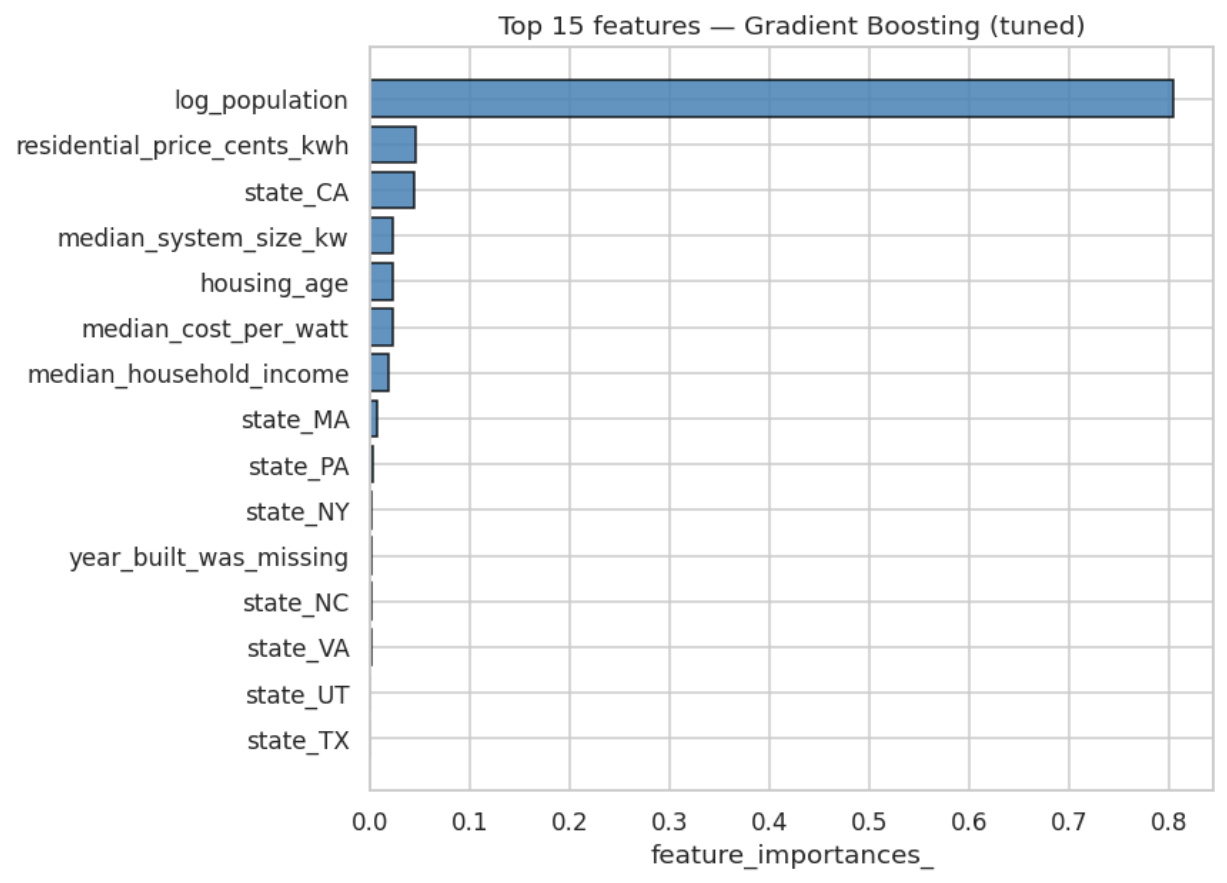
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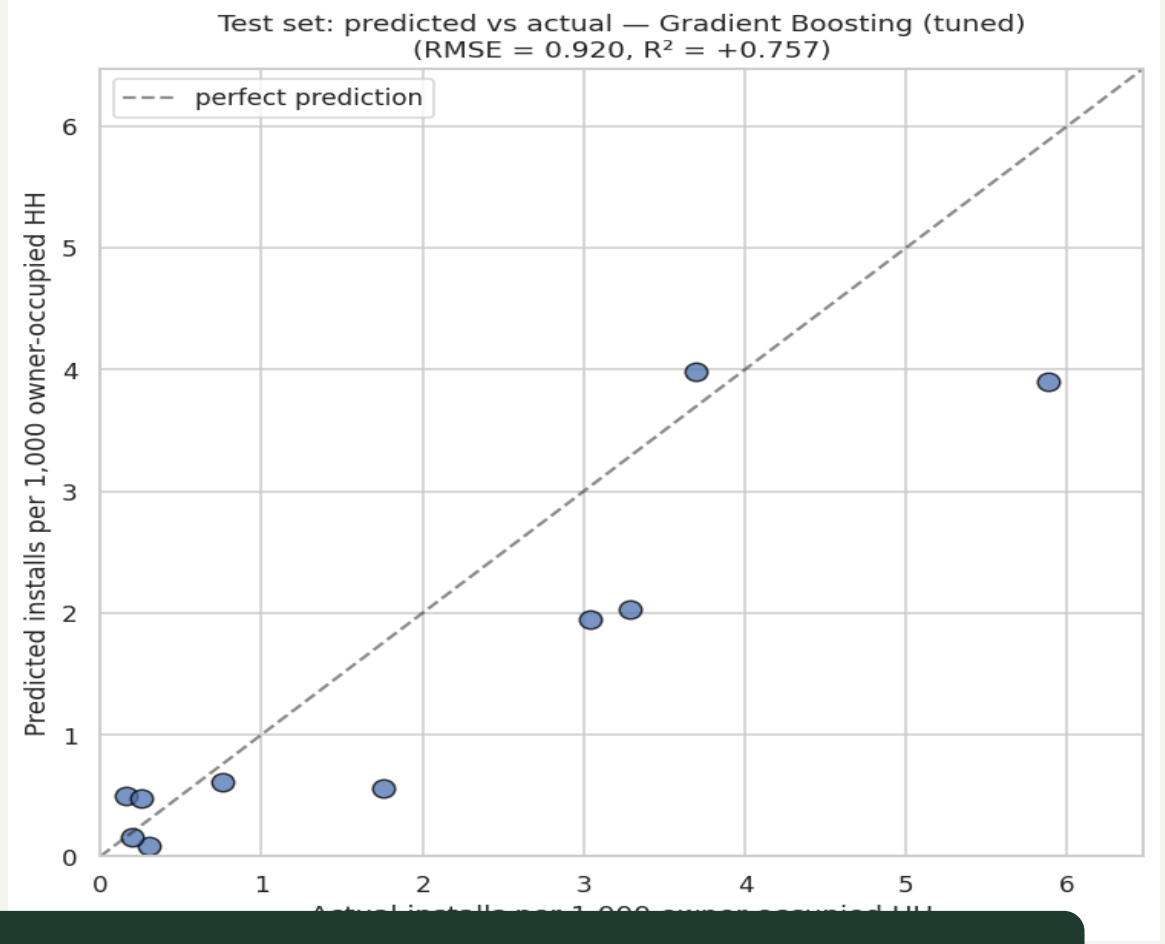
Both tree ensembles beat the linear models. Gradient Boosting wins by a comfortable margin — exactly the model architecture the project proposal committed to for the final version.

0.63	+0.28	0.92	+0.76
Cross-Val RMSE	Cross-Val R²	Test RMSE	Test R²

What drives the model's predictions?

Left: feature importance from the final Gradient Boosting model. Right: predicted vs. actual on the held-out test set.





log_population = 80% importance · residential electricity price is #2 · Test R^2 = **+0.76**

Three recommendations

1

SOLAR INSTALLERS

Use model residuals to find underserved markets

Counties where actual adoption is running below the model's prediction are underserved high-propensity markets. The residuals produce a rankable target list for market entry.

2

UTILITIES

Plan for distributed solar in mid-sized service areas

Population is negatively correlated with per-capita adoption. Feeder-level planning should prioritize mid-sized county service areas with elevated electricity prices — not the densest urban areas.

3

STATE ENERGY OFFICES

Reconsider income-targeted incentives

Income shows essentially no independent linear signal ($r \approx +0.06$). Price and housing stock look like stronger levers. Next incentive-program evaluation should test price- or tenure-based targeting.

Limitations & next steps

What's here works. What's next makes it deployable.

LIMITATIONS

Sample size. 50 counties in demo; ~3,100 available on full dataset. Confidence intervals will tighten dramatically.

Missing features. NREL irradiance and DSIRE policy indicators deferred to follow-up. Both named in the proposal.

Cross-sectional only. Single most-recent-year snapshot. Panel model would let us study within-county change.

Tenure proxy. Population is likely a proxy for housing tenure. Explicit tenure/unit-type feature would separate effects.

NEXT STEPS

1. **Run on full dataset**

   Same pipeline against ~3,100 counties nationwide.

2. **Add irradiance + policy features**

   NSRDB + DSIRE indicators, then re-tune.

3. **Deliver to stakeholders**

   Residual-ranked target lists for the three audiences.

Thank you.